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HT Board Booster

Class X Science

Physics

CBSE Board Paper Question Bank

Chapter Wise Questions

Ch - 1 Light

Session 2026-27



Class X Science Physics
Ch- Light
Previous Year CBSE Board Question Paper

CBSE BP 2025-26
(Set- 31/1/1- 31/5/1)
Section-A (one Marker Questions)

Q.1. A convex lens of focal length 15 cm, is forming a real image. If the size of image is same as the size of object, then position of object and position of image will be, respectively: (Set- 31/1/1)

(a) - 15 cm and - 15 cm from lens (b) - 15 cm and + 15 cm from lens
(c) - 30 cm and + 30 cm from lens (d) - 30 cm and - 30 cm from lens

Q.2. From the following options choose the one which gives negative magnification greater than one : (Set- 31/2/1)

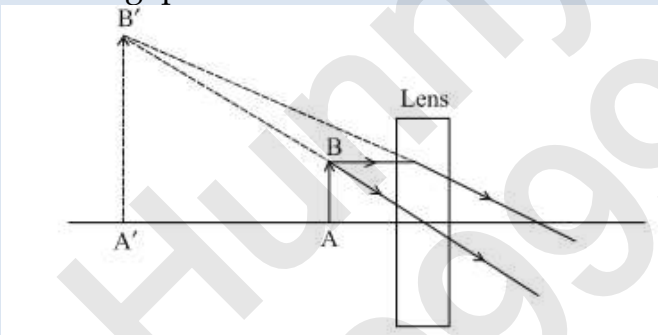
(a) Concave mirror; Nature of image - real
(b) Concave mirror; Nature of image - virtual
(c) Convex mirror; Nature of image - real
(d) Convex mirror; Nature of image – virtual

Q.3. Rays from the sun converge at a point 25 cm behind a convex lens. The distance at which an object be placed in front of the lens to get a virtual image, is: (Set- 31/5/1)

(a) 20 cm (b) 40 cm (c) 50 cm (d) More than 50 cm

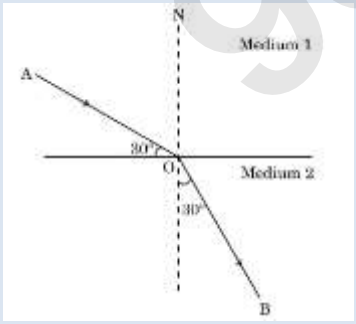
Section-B (two Marker Questions)

Q.4. The given figure shows image formation by a lens. Analyse the figure and answer the following questions: (Set- 31/1/1)



(a) What is the type of lens used for image formation in the given ray diagram
(b) If the real image is formed at a distance of 30 cm from the lens and the size of image is twice the size of the object, then where was the object placed?

Q.5.



With the help of the given ray diagram, calculate the refractive index of Medium 2 with respect to Medium 1. (Set- 31/2/1)

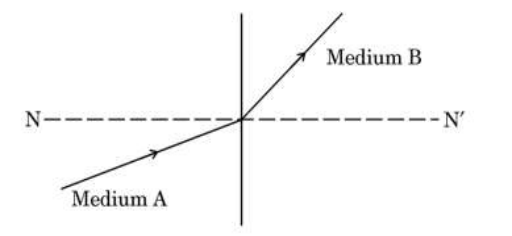
(Given that $\sqrt{3}$ = 1. 732)

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Q.6. (a) Write the expression for the magnification produced by a lens in terms of object distance and image distance. (Set- 31/2/1)

(b) A 4 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 20 cm. Calculate the size of the image formed, if the distance of the object from the lens is 10 cm.

Q.7.



A ray of light enters into medium B from medium A as shown in the given figure:

(a) Which one of these two media is optically denser than the other medium? Justify your answer. (Set- 31/3/1)

(b) If the speed of light in medium A is VA and in medium B is VB, then what is the refractive index of medium B with respect to medium A?

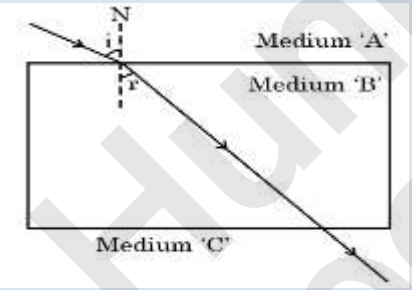
Q.8. (a) Relate the speed of light in the given medium with its optical density. (Set- 31/5/1)

(b) Using the information given in the table below, arrange the medium A, B and C in the ascending order of their optical density.

Medium	Speed of light (m/s)
A	2.25×10^8
B	2×10^8
C	2.08×10^8

Section-C (three Marker Questions)

Q.9.



(a) Observe the given ray diagram and compare : (Set- 31/4/1)

(i) the speed of light in medium A and the speed of light in medium B.

(ii) the optical densities of medium B and medium C.

(b) Draw a ray diagram for refraction where the optical density of medium 'A' becomes equal to the optical density of medium 'C' and optical density of medium 'B' is greater than the optical density of medium 'A' and medium 'C'.

Q.10. (a) A ray of light after reflection from a concave mirror becomes parallel to its principal axis. Draw a labelled ray diagram to show this. (Set- 31/5/1)

(b) (i) Differentiate between the virtual images formed by a convex mirror and concave mirror.

(ii) "The value of magnification 'm' for a mirror is-2." Calculate the height of object if the height of image is 20 cm.

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Section-D (four Marker Questions)

Q.11. Read the following passage and answer the questions given:

Lenses can form different types of images depending upon their focal length and position of object. A convex lens can create real, inverted or virtual, erect images, while a concave lens forms only virtual and diminished images. The focal length determines the power of lens. Convex lenses have positive focal length while concave lenses have negative focal length by convention. When lenses are placed together, their combined power is determined by the sum of their individual powers. Ray diagrams help to visualize how light converges or diverges through lens to form an image. (Set- 31/1/1)

(a) A convex lens of focal length 20 cm is used to form an image. If an object is placed at 40 cm from the lens, what will be the position and nature of image?

(b) Illustrate the formation of image with the help of ray diagram, when the object is placed between the optical centre and principal focus of concave lens.

(c) (i) A lens combination consists of a convex lens of focal length 30 cm and a concave lens of focal length 15 cm placed together. Find the equivalent focal length and power of this lens combination.

OR

(c) (ii) Two lenses are placed in contact. One is a concave lens with focal length 2 m and the other is a convex lens with focal length 1.5 m. What type of lens will the combination behave as (convex or concave)? Give reason.

Q.12. The variation of image distance (v) with the object distance (u) for a convex lens is given in the following observation table. Analyse it and answer the questions given below:

(Set- 31/3/1)

S.No.	Object distance (u) cm	Image distance (v) cm
1	– 150	+ 30
2	– 75	+ 37·5
3	– 50	+ 50
4	– 37·5	+ 75
5	– 30	+ 150
6	– 15	+ 37·5

(a) Without calculation, find out the focal length of the given convex lens. Justify your answer.

(b) Which one of the observations given in the table is not correct and why?

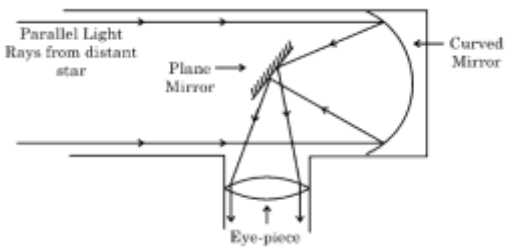
(c) (i) Find out the value of magnification for u = - 30 cm. Write the nature of the image formed. Give reason for your answer.

OR

(c) (ii) Analyse the data given in the table above and draw the labelled ray diagram for u = - 30 cm and v = 150 cm.

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Q.13.



The above ray diagram is to show image formation by a reflecting telescope. Reflecting telescope revolutionized our way of looking into the sky. Reflecting telescopes employ large mirrors to gather and focus light rather than relying solely on lenses as we do in the refracting telescopes. These telescopes utilize precisely shaped and polished mirrors to capture incoming light and reflect it to a focal point, where the telescope forms an image for observation. (Set- 31/4/1)

(a) What kind of image of a star is seen by the observer in the eye-piece?

(b) What kind of mirror is used in this reflecting telescope?

(c) Explain with the reason, what kind of optical device (type of lens or mirror) is used at the eye-piece.

OR

(c) What is the role of plane mirror in image formation by this telescope?

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CBSE BP 2024-25

Section-A (one Marker Questions)

Q.14. Mirror 'X' is used to concentrate sunlight in solar furnace and Mirror 'Y' is fitted on the side of the vehicle to see the traffic behind the driver. (Set- 31/1/1)

Which of the following statements are true for the two mirrors?

(i) The image formed by mirror 'X' is real, diminished and at its focus.
(ii) The image formed by mirror 'Y' is virtual, diminished and erect.
(iii) The image formed by mirror 'X' is virtual, diminished and erect.
(iv) The image formed by mirror 'Y' is real, diminished and at its focus.

(a) (i) and (ii) (b) (ii) and (iii) (c) (iii) and (iv) (d) (i) and (iv)

Q.15. If the absolute refractive indices of two media X and Y are and respectively, then the refractive index of Y with respect to X will be: 4/3 6/5 (Set- 31/2/1)

(a) 10/9 (b) 9/10 (c) 9/8 (d) 8/9

Q.16. An object is placed at a distance of 30 cm from the pole of a concave mirror. If its real and inverted image is formed at 60 cm in front of the mirror, the focal length of the mirror is:

(a) 15 cm (b) 20 cm (c) + 20 cm (d) + 15 cm (Set- 31/2/1)

Q.17. Absolute refractive index of water and glass is and $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m / s the speed of light in water is: (Set- 31/3/1)

(a) $\frac{9}{4}$ m / s (b) $\frac{7}{3}$ m / s (c) $\frac{16}{9}$ m / s (d) $\frac{9}{8}$ m / s

Q.18. A wire of length 'l' is gradually stretched so that its length increases to 31. If its original resistance is R, then its new resistance will be: (Set- 31/3/1)

(a) 3R (b) 6R (c) 9R (d) 27R

Q.19. To get an image of magnification -1 on a screen using a lens of focal length 20 cm, the object distance must be: (Set- 31/4/1)

(a) Less than 20 cm (b) 30 cm (c) 40 cm (d) 80 cm

Q.20. An optical device 'X' is placed obliquely in the path of a narrow parallel beam of light. If the emergent beam gets displaced laterally, the device 'X' is: (Set- 31/4/1)

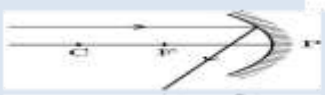
(a) plane mirror (b) convex lens (c) glass slab (d) glass prism

Q.21. In order to obtain large images of the teeth of patients, the dentist holds the concave mirror in such a manner that the teeth are positioned (Set- 31/5/1)

(a) at the focus of mirror.
(b) between pole and focus of the mirror.
(c) between focus and centre of curvature of the mirror.
(d) at the centre of curvature of the mirror.

Q.22. Identify from the following the ray diagram which shows the correct path of the reflected ray for the the ray incident on a concave mirror as shown:



(A)


(B)


(C)


(D)


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Section-B (two Marker Questions)

Q.23. An object is placed at a distance of 60 cm from a concave lens of focal length 30 cm. Use lens formula to find the position of the image formed in this case. (Set- 31/1/1)

Q.24. (a) Out of the two lenses, one concave and the other convex, state which one will diverge a parallel beam of light falling on it. Draw a ray diagram to show the principal focus of the lens. (Set- 31/2/1)

OR

(b) A ray of light after refraction from a convex lens emerges parallel to its principal axis.
(i) Draw a labelled ray diagram to show it.
(ii) In this case, the incident ray before refraction from the lens passes through a point on its principal axis. Name the point.

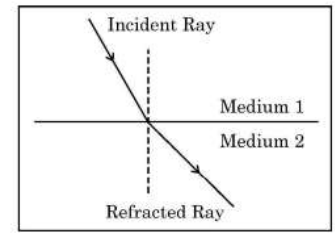
Q.25. Draw a ray diagram to show the refraction of a ray of light passing through an equilateral glass prism. Mark the angle through which the emergent ray bends from the direction of the incident ray and also name it. (Set- 31/4/1)

OR

Name the type of lenses required by the persons for the correction of their defect of vision called presbyopia. Write the structure of the lenses commonly used for the correction of this defect giving reason for such designs.

Q.26. The values of absolute refractive indices of kerosene and water are 1.44 and 1.33 respectively. Compare the two media on the basis of their (Set- 31/5/1)
(a) optical density
(b) mass density
(c) relative speed of propagation of light.
What do you infer on the basis of above comparisons?

Q.27. Study the figure in which the path of a ray of light going from Medium 1 to Medium 2 is shown. (Set- 31/6/1)



(a) Out of the two Media - Medium 1 and Medium 2, in which is the speed of light more?
(b) State reason of bending of the refracted ray away from the normal.
(c) Express refractive index of Medium 2 with respect to Medium 1 in terms of speed of light in two media.

Section-C (three Marker Questions)

Q.28. Draw ray diagrams to show the nature, position and relative size of the image formed by a convex mirror when the object is placed (i) at infinity and (ii) between infinity and pole P of the mirror. (Set- 31/1/1)

Q.29. A convex lens forms an 8-0 cm long image of a 2-0 cm long object which is kept at a distance of 6-0 cm from the optical centre of the lens. If the object and the image are on the same side of the lens, find (i) the nature of the image, (ii) the position of the image, and (iii) the focal length of the lens. (Set- 31/2/1)

Q.30. What is a rainbow? Draw a labelled diagram to show its formation. (Set- 31/3/1)

Q.31. If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer. (Set- 31/4/1)

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Q.32. The power of a lens is -0.25 D. Based on this information, find out (Set- 31/5/1)

(a) The type of lens and its focal length.

(b) The eye defect for which it may be used as a corrective lens.

(c) The nature and size of the image formed by this lens when an object is placed between F and 2F from the optical centre of this lens.

Q.33. A student placed a candle flame at different distances from a convex lens and focused its image on a screen. He recorded his observation in tabular form as given below: (Set- 31/6/1)

S.No.	Distance of flame from the lens (cm)	Distance of the image from the lens
1	– 90	+ 18
2	– 60	+ 20
3	– 40	+ 24
4	– 30	+ 30
5	– 24	+ 40
6	– 20	+ 60
7	– 18	+ 90
8	– 12	+ 120

Analyse the observation table and on the basis of your analysis only, answer the following questions (without doing any calculations) :

(a) What is the focal length of the convex lens used ? Give reason to justify your answer.

(b) Which one of the sets of observations is not correct and why?

(c) Draw ray diagram to show image formation for any correct set of observation.

Section D

Q.34. (i) The power of a lens 'X' is 2.5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens? (Set- 31/1/1)

(ii) "The value of magnification 'm' for a lens is 2." Using new Cartesian Sign Convention and considering that an object is placed at a distance of 20 cm from the optical centre of this lens, state:

(I) the nature of the image formed;

(II) size of the image compared to the size of the object;

(III) position of the image, and (IV) sign of the height of the image.

(iii) The numerical values of the focal lengths of two lenses A and B are 10 cm and 20 cm respectively. Which one of the two will show higher degree of convergence/divergence? Give reason to justify your answer.

OR

(i) Draw a ray diagram to show the refraction of a ray of light through a rectangular glass slab when it falls obliquely from air into glass.

(ii) State Snell's law of refraction of light.

(iii) Differentiate between the virtual images formed by a convex lens and a concave lens on the basis of:

(I) object distance, and (II) magnification.

Q.35. (i) "In refraction of light through a rectangular glass slab, the emergent ray is always

parallel to the direction of the incident ray." Why? Explain with the help of a ray diagram. What happens when a ray of light falls normally on one of the faces of a rectangular glass prism? Draw diagram. (Set- 31/3/1)

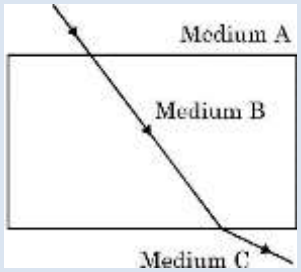
(ii) An object is placed at a distance of 30 cm from the optical centre of a concave lens of focal length 20 cm. Use Lens formula to determine the position of the image formed in this case.

OR

(i) A student wishes to study the image formation by a concave mirror using candle flame as object. State the type of the image formed by the mirror and mention the change in the image formed, if any, that he observes when the candle flame is gradually moved away from the pole of the mirror. Draw a ray diagram to show the image formation when the object distance is nearly equal to the radius of curvature of the mirror.

(ii) A convex mirror used for rear-view on an automobile has a focal length of 3-0 m. If a bus is located at 6-0 m from this mirror, use mirror formula to find the position of the image of the bus as seen in the mirror.

Q.36. (A) (a) Observe the following diagram and compare (i) speed of light and (ii) optical densities of the three media A, B and C. Also give justification for your answer of any one of the two cases in terms of refractive indices of A, B and C. (Set- 31/5/1)



(b) Redraw the path of a ray of light through the three media, if the ray of light starting from medium A falls on the medium B

(i) Obliquely and the optical density of medium B is made more than that of A and C.

(ii) The ray falls normally from medium A to medium B.

OR

(B) Analyse the following observation table showing variation of image distance (v) with object distance (u) in case of a convex lens and answer the questions that follow without doing any calculations:

Observation Number	Object distance (u) in cm	Image Distance (v) in cm
1	- 15	- 60
2	- 25	+ 100
3	- 30	+ 60
4	- 40	+ 40
5	- 60	+ 30
6	- 100	+ 25

(a) Determine the focal length of the lens. Give reason for your answer.

(b) Find magnification of the image formed in Observation No. 3.

(c) The numerical value of magnifications in cases of observation 1 and 2 is same. List two differences in the images formed in these two cases.

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Section-A (one Marker Questions)

Q.37. At what distance from a convex lens should an object be placed to get an image of the same size as that of the object on a screen? (Set- 31/1/1)

(a) Beyond twice the focal length of the lens.
(b) At the principal focus of the lens,
(c) At twice the focal length of the lens.
(d) Between the optical centre of the lens and its principal focus

Q.38. Absolute refractive index of glass and water is $\frac{3}{4}$ and $\frac{4}{3}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is: (Set- 31/4/1)

(A) $\frac{9}{4} \times 10^8$ m/s (B) $\frac{5}{2} \times 10^8$ m/s

(C) $\frac{7}{3} \times 10^8$ m/s (D) $\frac{16}{9} \times 10^8$ m/s

Q.39. How will the image formed by a convex lens be affected, if the upper half of the lens is wrapped with a black paper? (Set- 31/5/1)

(a) The size of the image formed will be one-half of the size of the image due to complete lens.
(b) The image of upper half of the object will not be formed.
(c) The brightness of the image will reduce.
(d) The lower half of the inverted image will not be formed.

Section-B (two Marker Questions)

Q.40. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position of the image formed by the mirror. (Set- 31/1/1)

Q.41. State two laws of refraction of light. (Set- 31/2/1)

OR

Define the term absolute refractive index of a medium. A ray of light enters from vacuum to glass of absolute refractive index 1.5. Find the speed of light in glass. The speed of light in vacuum is 3 x 108 m/s.

Q.42. The heat produced at a point due to concentration of sunlight by a convex lens burns a paper.

(a) Explain why it happens. (Set- 31/5/1)
(b) Name the term (in the context of the lens used) given to the point at which the paper starts burning. What does the bright spot formed on the paper represent ?

Section –D

Q.43. (a) Upper half of a convex lens is covered with a black paper. Draw a ray diagram to show the formation of image of an object placed at a distance of 2F from such a lens. Mention the position and nature of the image formed. State the observable difference in the image obtained if the lens is uncovered. Give reason to justify your answer.

(b) An object is placed at a distance of 30 cm from the optical centre of a concave lens of focal length 15 cm. Use lens formula to determine the distance of the image from the optical centre of the lens. (Set- 31/2/1)

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Q.44. Draw a ray diagram to show the path of the refracted ray in each of the following cases:
A ray of light incident on a concave lens (Set- 31/3/1)

(1) parallel to its principal axis, and

(2) is directed towards its principal focus.

(ii) A 4 cm tall object is placed perpendicular to the principal axis of convex lens of focal length 24 cm. The distance of object from the lens is 16 cm. Find the position and size of image formed.

OR

Draw a ray diagram to show the path of the reflected ray in each of the following cases:
A ray of light incident on a convex mirror

(1) parallel to its principal axis, and

(2) is directed towards its principal focus

(ii) A 1.5 cm tall candle flame is placed perpendicular to the principal axis of a concave mirror of focal length 12 cm. If the distance of the flame from the pole of the mirror is 18 cm, use mirror formula to determine the position and size of the image formed.

Q.45. The variation of image distance (v) with object distance (u) for a convex lens is given in the following observation table. Analyse it and answer the questions that follow :

S.NO.	Object Distance (u) cm	Image Distance (v) cm
1	-150	+30
2	-75	+37.5
3	-50	+50
4	37.5	+75
5	-30	+150
6	-15	+37.5

(i) Without calculation, find the focal length of the convex lens. Justify your answer.

(ii) Which observation is not correct ? Why ? Draw ray diagram to find the position of the image formed for this position of the object.

(iii) Find the approximate value of magnification for u = 30 cm.

(b) (i) Define principal axis of a lens. Draw a ray diagram to show what happens when a ray of light parallel to the principal axis of a concave lens passes through it.

(ii) The focal length of a concave lens is 20 cm. At what distance from the lens should a 5 cm tall object be placed so that its image is formed at a distance of 15 cm from the lens ? Also calculate the size of the image formed. (Set- 31/4/1)

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Section – E

Q.46. Study the data given below showing the focal length of three concave mirrors A, B and C and the respective distances of objects placed in front of the mirrors: (Set- 31/1/1)

Case	Mirror	Focal length (cm)	Object Distance (cm)
1	A	20	45
2	B	15	30
3	C	30	20

(i) In which one of the above cases the mirror will form a diminished image of the object? Justify your answer.

(ii) List two properties of the image formed in case 2.

(iii) (A) What is the nature and size of the image formed by mirror C? Draw ray diagram to justify your answer.

OR

(iii) (B) An object is placed at a distance of 18 cm form the pole of a concave mirror of focal length 12cm. Find the position of the image formed in this case.

Q.47. A highly polished surface such as a mirror reflects most of the light falling on it. In our daily life we use two types of mirrors plane and spherical. The reflecting surface of a spherical mirrors may be curved inwards or outwards. In concave mirrors, reflection takes place from the inner surface, while in convex mirrors reflection takes place from the outer surface.

(a) Define the principal axis of a concave mirror.

(b) A ray of light is incident on a concave mirror, parallel to its principal axis. If this ray after reflection from the mirror passes through the principal axis from a .point at a distance of 10 cm from the pole of the mirror, find the radius of curvature Of the mirror.

(c) (i) An object is placed at a distance of 10 cm from the pole of a convex mirror of focal length 15 cm. Find the position of the image.

OR

(c) (ii) A mirror forms a virtual, erect and diminished image of an object. Identify the type of this mirror. Draw a ray diagram to show the image formation in this case.

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CBSE BP 2022-23

Section-A (one Marker Questions)

Q.48. An object is placed in front of a convex mirror at infinity. According to the New Cartesian Sign Convention, the sign of the focal length and the sign of the image distance in this case are respectively: (Set- 31/2/1)

(a) +, - (b) -, + (c) - , - (d) +,+

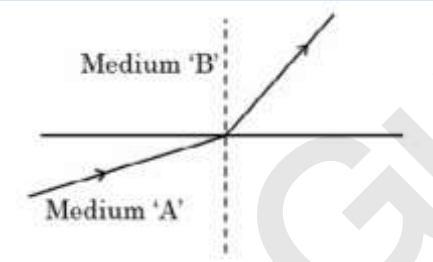
Q.49. A student wants to obtain an erect image of an object using a concave mirror of 10 cm focal length. What will be the distance of the object from mirror? (Set- 31/6/1)

(a) Less than 10 cm (b) 10 cm

(c) between 10 cm and 20 cm (d) more than 20 cm

Section-B (two Marker Questions)

Q.50. A light ray enters from medium A to medium B as shown in the figure. (Set- 31/6/1)



(a) Which one of the two media is denser w.r.t. other medium? Justify your answer.

(b) If the speed of light in medium A is u, and in medium B is the refractive index of B with respect to A. what is

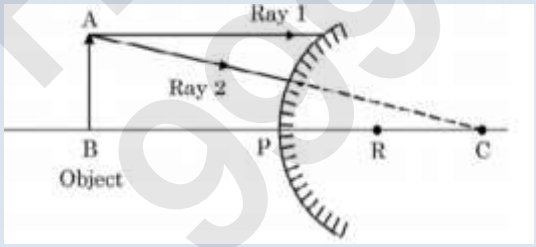
OR

(a) A ray of light starting from diamond is incident on the interface separating diamond and water. Draw a labelled ray diagram to show the refraction of light in this case.

(b) Absolute refractive indices of diamond and water are 2.42 and 1.33 respectively. Find the value of refractive index of water w.r.t. diamond.

Section-C (three Marker Questions)

Q.51. (a) Complete the following ray diagram to show the formation of image: (Set- 31/1/1)



(b) Mention the nature, position and size of the image formed in this case.

(c) State the sign of the image distance in this case using the Cartesian sign convention.

Q.52. (a) A student has focussed the image of an object of height 3 cm on a white screen using

A concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following: (Set- 31/2/1)

(i) Distance of the image from the mirror

(ii) Height of the image

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OR

(b) Define power of a lens. The focal length of a lens is - 10 cm. Write the nature of the lens and find its power. If an object is placed at a distance of 20 cm from the optical centre of this lens,
according to the New Cartesian Sign Convention, what will be the sign of magnification in this case?

Q.53. (a) Define the following terms in the context of a diverging mirror: (Set- 31/4/1)
(i) Principal focus (ii) Focal length
Draw a labelled ray diagram to illustrate your answer.

(b) An object of height 10 cm is placed 25 cm away from the optical centre of a converging lens of focal length 15 cm. Calculate the image-distance and height of the image formed.

Q.54. The power of a lens is +4D. Find the focal length of this lens. An object is placed at a distance of 50 cm from the optical centre of this lens. State the nature and magnification of the image formed by the lens and also draw a ray diagram to justify your answer. (Set- 31/4/1)

Q.55. (a) (i) An object of 5 cm height is placed at a distance of 20 cm from the optical centre of a concave lens of focal length 18 cm. Calculate (1) image distance and (2) the magnification in this case. (Set- 31/5/1)

(ii) Compare the values of magnification obtained by a concave lens and a convex lens when both the lenses form virtual images.

OR

(b) A convex lens can form a (i) real, inverted and magnified image as well as (ii) virtual, erect and magnified image of an object. If the focal length of the lens is 10 cm, what should be the range of the object distance in both cases? Draw ray diagrams to justify Your answer.

Section-D

Q.56. (a) An object is placed in front of a convex lens of focal length f. If the distance of the object from the lens is 2f, draw a ray diagram to show the formation of the image. Write the value of magnification in this case. (Set- 31/2/1)

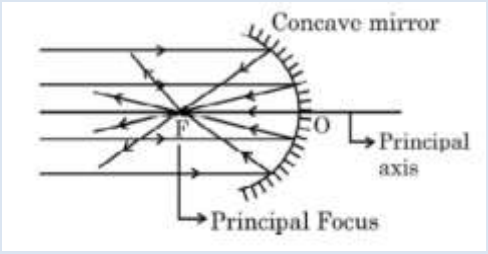
(b) A student has focussed the image of a candle flame on a white screen using a convex lens. The situation is as given below:
Length of the flame = 2 cm
Focal length of the lens = 12 cm
Distance of the flame from the lens = 16 cm
If the flame is perpendicular to the principal axis of the lens, calculate the values of the following:

(i) Distance of the image from the lens
(ii) Length of the image formed

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Section-E

Q.57. Hold a concave mirror in your hand and direct its reflecting surface towards the sun. Direct the light reflected by the mirror on to a white card-board held close to the mirror. Move the card-board back and forth gradually until you find a bright, sharp spot of light on the board. This spot of light is the image of the sun on the sheet of paper; which is also termed as "Principal Focus" of the concave mirror.



(a) List two applications of concave mirror. (Set- 31/1/1)

(b) If the distance between the mirror and the principal focus is 15 cm, find the radius of curvature of the mirror.

(c) Draw a ray diagram to show the type of image formed when an object is placed between Pole and focus of a concave mirror.

OR

(c) An object 10 cm in size is placed at 100 cm in front of a concave mirror. If its image is formed at the same point where the object is located, find:

(i) Focal length of the mirror, and

(ii) Magnification of the image formed with sign as per Cartesian sign convention.

Q.58. The ability of a medium to refract light is expressed in terms of its optical density. Optical density has a definite connotation. It is not the same as mass density. On comparing two media, the one with the large refractive index is optically denser medium than the other. The other medium with a lower refractive index is optically rarer. Also the speed of light through a given medium is inversely proportional to its optical density. (Set- 31/4/1)

(i) Determine the speed of light in diamond if the refractive index of diamond with respect to vacuum is 2.42. Speed of light in vacuum is 3×108 m/s.

(ii) Refractive indices of glass, water and carbon disulphide are 1.5, 1.33 and 1.62 respectively. If a ray of light is incident in these media at the same angle (say 0), then write the increasing order of the angle of refraction in these media.

(iii) (A) The speed of light in glass is 2×108 m/s and in water is 2.25×10 m/s.

(a) Which one of the two is optically denser and why?

(b) A ray of light is incident normally at the water-glass interface when it enters a thick glass container filled with water. What will happen to the path of the ray after entering the glass? Give reason.

OR

(iii) (B) The absolute refractive indices of water and glass are 4/3 and 3/2 respectively. If the speed of light in glass is 2×108 m/s, find the speed of light in (i) vacuum and (ii) water

Q.59. Many optical instruments consist of a number of lenses. They are combined to increase the magnification and sharpness of the image. The net power (P) of the lenses placed in contact is given by the algebraic sum of the powers of the individual lenses P1, P2, P3... as P=P1+ P2+P3... (Set- 31/5/1)

This is also termed as the simple additive property of the power of lens, widely used to design lens systems of cameras, microscopes and telescopes. These lens systems can have a combination of convex lenses and also concave lenses.

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(a) What is the nature (convergent / divergent) of the combination of a convex lens of power + 4 D and a concave lens of power - 2 D?

(b) Calculate the focal length of a lens of power - 2.5 D.

(c) Draw a ray diagram to show the nature and position of an image formed by a convex lens of power + 0.1 D, when an object is placed at a distance of 20 cm from its optical centre.

OR

(c) How is a virtual image formed by a convex lens different from that formed by a concave lens? Under what conditions do a convex and a concave lens form virtual images?

Q.60. A student took three concave mirrors of different focal lengths and performed the experiment to see the image formation by placing an object at different distances with these mirrors as shown in the following table. (Set- 31/6/1)

Case No.	Object-distance	Focal length
I	45 cm	20 cm
II	30 cm	15 cm
III	20 cm	30 cm

Now answer the following questions:

(a) List two properties of the image formed in Case I.

(b) In which one of the cases given in the table, the mirror will form real image of same size and why?

(c) Name the type of mirror used by dentists. Give reason why do they use such type of mirrors.

OR

(c) Look at the table and identify the situation (object distance and focal length) which resembles the situation in which concave mirrors are used as shaving mirrors? Draw a ray diagram to show the image formation in this case.

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